MAT1001 Calculus for Engineers

Gamma and Beta Functions and Applications of Integration

Dr. Aswathy R K
Department of Mathematics
School of Advanced Sciences
Vellore Institute of Technology – Andhra Pradesh

VIT-AP UNIVERSITY

Gamma function

The gamma function of n is defined as

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$$

Properties:

- $\Gamma(1) = 1$
- $\Gamma(n+1) = n\Gamma(n)$
- $\Gamma(n+1) = n!$

Transformation of Gamma function

: (1) $\int_0^\infty e^{-ky} y^{n-1} dy = \frac{\Gamma n}{k^n}$ (2) $\Gamma \frac{1}{2} = \sqrt{\pi}$ (3) $\int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dy = \Gamma n$

VIT-AP UNIVERSITY

Problem

Find the area under curve $x^{1/4}e^{-\sqrt{x}}$ for $0 \le x < \infty$

Solution. Let $I = \int_0^\infty x^{1/4}\, e^{-\sqrt{x}}\, dx$

Putting $\sqrt{x} = t$ or $x = t^2$ or $dx = 2\,t\,dt$ in (1), we get

$$I = \int_0^\infty t^{1/2}\, e^{-t}\, 2\,t\,dt = 2\int_0^\infty t^{3/2}\, e^{-t}\, dt$$

$$= 2\,\Gamma\frac{5}{2} \qquad \text{By definition}$$

$$= 2\,.\,\frac{3}{2}\,\Gamma\frac{3}{2} = 2\,.\,\frac{3}{2}\,.\,\frac{1}{2}\,\Gamma\frac{1}{2} = \frac{3}{2}\sqrt{\pi}$$

Excercise

Find the area under the curve $\sqrt{x}e^{-\sqrt[3]{x}}$ for $0 \le x < \infty$

Beta Function

VIT-AP UNIVERSITY

Beta function

Beta function of l, m is defined as

$$\beta(l,m) = \int_0^1 x^{l-1}(1-x)^{m-1}dx \qquad (l>0, m>0)$$

Symmetric property of beta function

$$\beta(l,m) = \beta(m,l)$$

Relation between beta and gamma function

$$\beta(l,m) = \frac{\Gamma(l)\Gamma(m)}{\Gamma(l+m)}$$

Example

Problem

Find the area under the curve $x^4(1-\sqrt{x})^5$ and $0 < x < 1$

Solution. Let $\sqrt{x} = t$ or $x = t^2$ or $dx = 2\,t dt$

$$\int_0^1 x^4 (1-\sqrt{x})^5\, dx = \int_0^1 (t^2)^4 (1-t)^5 (2\, t dt)$$

$$= 2\int_0^1 t^9 (1-t)^5\, dt = 2\,\beta(10, 6) = 2\frac{\Gamma 10\,\Gamma 6}{\Gamma 16} = 2\frac{\lfloor 9 \lfloor 5}{\lfloor 15}$$

$$= 2\,.\,\frac{\lfloor 5}{10\times 11\times 12\times 13\times 14\times 15} = \frac{2\times 1\times 2\times 3\times 4\times 5}{10\times 11\times 12\times 13\times 14\times 15}$$

$$= \frac{1}{11\times 13\times 7\times 15} = \frac{1}{15015}$$

Excercise

Find the area under the curve $(1-x^3)^{-1/2}$ and $0 < x < 1$

VIT-AP UNIVERSITY

Another form of beta function $\beta(l,m)=\int_0^\infty \frac{x^{l-1}}{(1+x)^{m+l}}dx$

Example . *Evaluate* $\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx$

Solution. We know that

$$\beta(m,n)=\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx$$

$$\Rightarrow \quad \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}}dx+\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}}=\beta(m,n) \qquad \ldots(1)$$

$$\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx=\int_1^0 \frac{\left(\frac{1}{t}\right)^{m-1}}{\left(1+\frac{1}{t}\right)^{m+n}}\left(-\frac{1}{t^2}dt\right)=\int_0^1 \frac{\left(\frac{1}{t}\right)^{m-1}\frac{1}{t^2}}{\frac{(t+1)^{m+n}}{\left(\frac{1}{t}\right)^{m+n}}}dt \qquad \left(\text{Put } x=\frac{1}{t}\right)$$

$$=\int_0^1 \frac{t^{n-1}}{(1+t)^{m+n}}dt=\int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}}dx$$

Beta Function – Another Form (Continued)

Putting the value of $\int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx$ in (1) we get

$$\int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}}\,dx + \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}}\,dx = \beta(m, n)$$

$$\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}\,dx = \beta(m, n)$$

Some Extra Results

- $\int_0^{\pi/2} \sin^p\theta \cos^q\theta = \frac{\Gamma(\frac{p+1}{2})\Gamma(\frac{q+1}{2})}{2\Gamma(\frac{p+q+2}{2})}$
- $\Gamma(m)\Gamma(m+1/2) = \frac{\sqrt{\pi}}{2^{2m-1}}\Gamma(2m)$
- $\beta(m,m) = 2^{1-2m}\beta(m,1/2)$

Applications : Mass of an Inhomogeneous Lamina

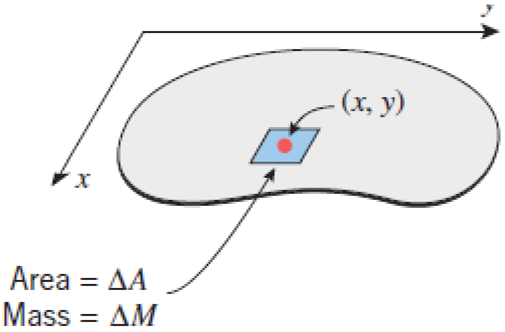
An idealized flat object that is thin enough to be viewed as a two-dimensional plane region is called a *lamina*

A lamina is called ***homogeneous*** if its composition is uniform throughout and ***inhomogeneous*** otherwise

The thickness of a lamina is negligible.

The *density* of a *homogeneous* lamina of mass M and area A is given by $\delta = M/A$

For an inhomogeneous lamina the density at a point (x, y) can be specified by a function $\delta(x, y)$, called the ***density function***

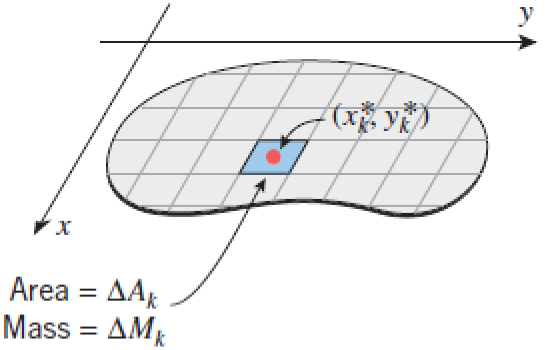


$$\delta(x, y) = \lim_{\Delta A \to 0} \frac{\Delta M}{\Delta A}$$

$$\Delta M \approx \delta(x, y)\Delta A$$

Mass of an Inhomogeneous Lamina

Imagine the lamina to be subdivided into rectangular pieces



Assume that there are n such rectangular pieces, and suppose that the kth piece has area ΔA_k

(x_k^*, y_k^*) denote the center of the kth piece.

the mass ΔM_k of this piece can be approximated by

$$\Delta M_k \approx \delta(x_k^*, y_k^*)\Delta A_k$$

and hence the mass M of the entire lamina can be approximated by $M \approx \sum_{k=1}^{n} \delta(x_k^*, y_k^*)\Delta A_k$

$$\boxed{M = \lim_{n \to +\infty} \sum_{k=1}^{n} \delta(x_k^*, y_k^*)\Delta A_k = \iint_R \delta(x, y)\, dA}$$

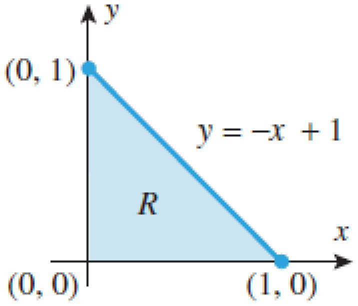
Mass of an Inhomogeneous Lamina

MASS OF A LAMINA If a lamina with a continuous density function $\delta(x, y)$ occupies a region R in the xy-plane, then its total mass M is given by

$$M = \iint_R \delta(x, y)\, dA$$

- A triangular lamina with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$ has density function $\delta(x, y) = xy$. Find its total mass.

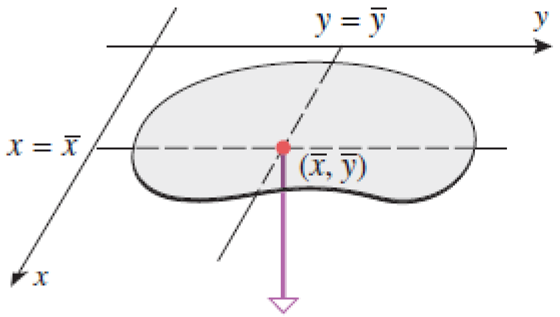
➡



$$M = \iint_R \delta(x, y)\, dA = \iint_R xy\, dA = \int_0^1 \int_0^{-x+1} xy\, dy\, dx$$

$$= \int_0^1 \left[\frac{1}{2}xy^2\right]_{y=0}^{-x+1} dx = \int_0^1 \left[\frac{1}{2}x^3 - x^2 + \frac{1}{2}x\right] dx = \frac{1}{24}$$

Center of Gravity of an Inhomogeneous Lamina



Center of Gravity $(\bar{x}, \bar{y})$ of a Lamina

$$\bar{x} = \frac{\displaystyle\iint_R x\delta(x, y)\, dA}{\displaystyle\iint_R \delta(x, y)\, dA}, \qquad \bar{y} = \frac{\displaystyle\iint_R y\delta(x, y)\, dA}{\displaystyle\iint_R \delta(x, y)\, dA}$$

Recall that if a point-mass m is located at the point (x, y), then the moment of m about $x = a$ measures the tendency of the mass to produce a rotation about the line $x = a$, and the moment of m about $y = c$ measures the tendency of the mass to produce a rotation about the line $y = c$. The moments are given by the following formulas:

$$M_x = \iint_R y\delta(x, y)\, dA \qquad M_y = \iint_R x\delta(x, y)\, dA$$

$$\bar{x} = \frac{M_y}{M} \quad \bar{y} = \frac{M_x}{M}$$

Example

- Find the center of gravity of the triangular lamina with vertices (0, 0), (0, 1), and (1, 0) and density function $\delta(x, y) = xy$.

➡

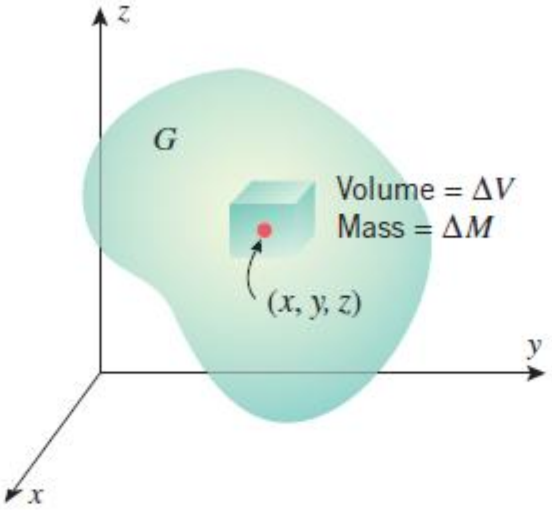
$$M = \iint_R \delta(x, y)\, dA = \iint_R xy\, dA = \frac{1}{24}$$

$$M_y = \iint_R x\delta(x, y)\, dA = \iint_R x^2 y\, dA = \int_0^1 \int_0^{-x+1} x^2 y\, dy\, dx = \frac{1}{60}$$

$$M_x = \iint_R y\delta(x, y)\, dA = \iint_R xy^2\, dA = \int_0^1 \int_0^{-x+1} xy^2\, dy\, dx = \frac{1}{60}$$

$$\bar{x} = \frac{M_y}{M} = \frac{1/60}{1/24} = \frac{2}{5}, \quad \bar{y} = \frac{M_x}{M} = \frac{1/60}{1/24} = \frac{2}{5}$$

Mass and Center of Gravity of a Solid



$$M = \text{mass of } G = \iiint_G \delta(x, y, z)\, dV$$

Center of Gravity $(\bar{x}, \bar{y}, \bar{z})$ *of a Solid G*

$$\bar{x} = \frac{1}{M} \iiint_G x\delta(x, y, z)\, dV$$

$$\bar{y} = \frac{1}{M} \iiint_G y\delta(x, y, z)\, dV$$

$$\bar{z} = \frac{1}{M} \iiint_G z\delta(x, y, z)\, dV$$

Example

- Find the mass and the center of gravity of a cylindrical solid of height h and radius a (Figure 14.8.8), assuming that the density at each point is proportional to the distance between the point and the base of the solid.

➡

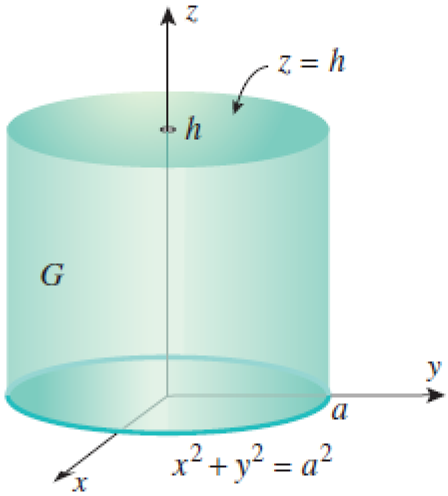


Figure 14.8.8

$$\delta(x, y, z) = kz$$

$$M = \iiint_G \delta(x, y, z)\, dV = \int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_0^h kz\, dz\, dy\, dx \quad = \tfrac{1}{2}kh^2\pi a^2$$

$$\bar{z} = \frac{1}{M} \iiint_G z\delta(x, y, z)\, dV = \frac{1}{\frac{1}{2}kh^2\pi a^2} \iiint_G z\delta(x, y, z)\, dV$$

$$= \frac{1}{\frac{1}{2}kh^2\pi a^2} \int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_0^h z(kz)\, dz\, dy\, dx \quad = \frac{2}{3}h$$

$$\bar{x} = \bar{y} = 0.$$

Thus, the center of gravity is $\left(0, 0, \frac{2}{3}h\right)$.

Thank You…